Unit 10 Surface Chemistry

S.No.	Answers		
1.	(c) $\frac{x}{m}=k.p^{1/n}$ $\Rightarrow \log(\frac{x}{m})=\log k+\frac{1}{n}\log p$ $y=c+mx$ $\quad m=\frac{1}{n}$ and $c=\log k$		
2.	The incorrect statement is option (b) Physisorption is an exothermic process. Hence increase in temperature decreases the physisorption.		
3.	(d) Adsorption leads to decrease in randomness (entropy).i.e. $\Delta S< 0$ for the adsorption to occur, ΔG should be -ve. We know that $\Delta G=\Delta H-T\Delta S$ if ΔS is -ve, $T\Delta S$ is +ve. It means that ΔG will become negative only when ΔH is -ve and $\Delta H>T\Delta S$		
4.	(c) dispersion medium-gas dispersed phase-liquid	13.	pyroxylin(nitro cellulose)
5.	(a) (Hardy-Schulze rule)	14.	(d) Both reactant and catalyst are in same phase. i.e(l)
6.	(b)	15.	(a)
7.	(b) Emulsion dispersed phase Dispersion medium -liquid	16.	(a) $\text{coagulating power } \alpha \frac{1}{\text{coagulation value}}$
8.	(b) Gel-butter	17.	(d) ΔS is -ve
9.	(d) As_2S_3 is a -vely charged colloid. It will be most effectively coagulated by the cation with greater valency. i.e., Al^{3+}.	18.	(d)
10.	(b)	19.	(a)
11.	(d) Tyndall effect-scattering of light	20.	(d)
12.	(b)		

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